Viney Washington

Unit 5 Project Submission

Exercise 1: Classify network services and applications according to protocol.

Figure 1: Lab Resources folder

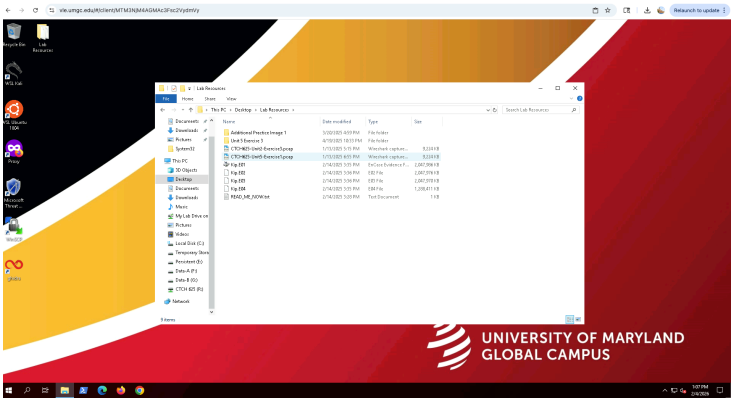


Figure 2: CTCH625-Unit5- Exercise 1. Cap

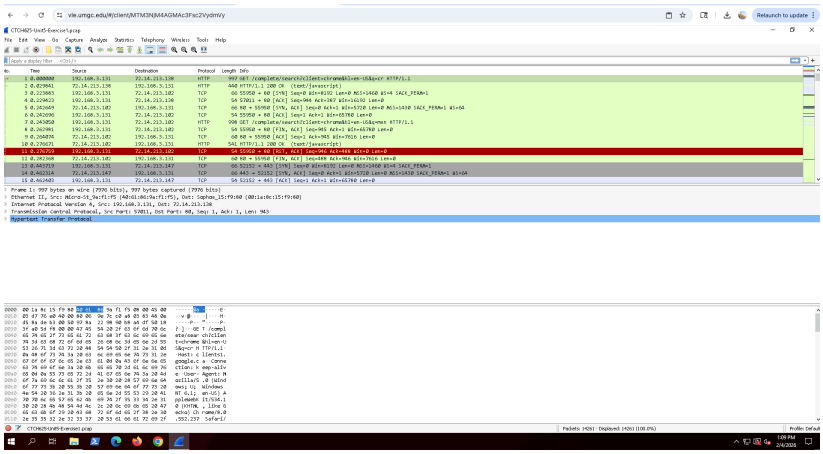
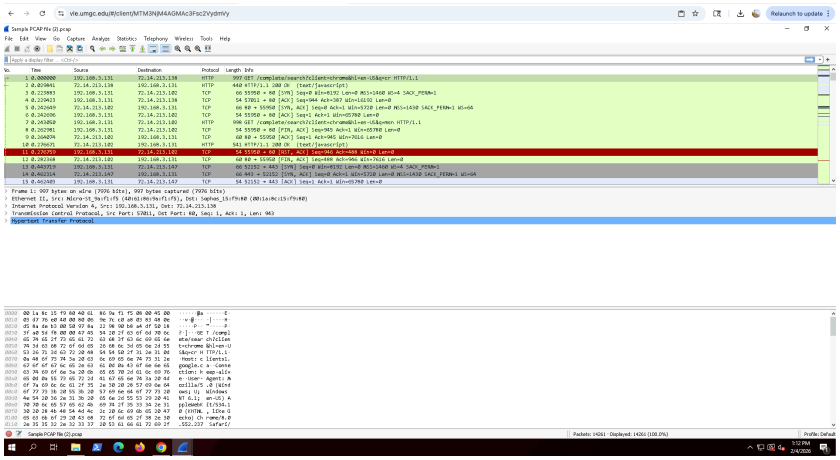
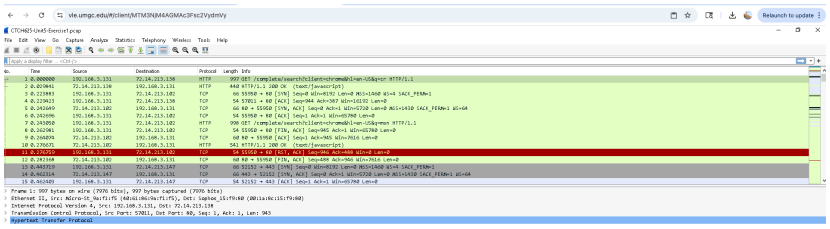
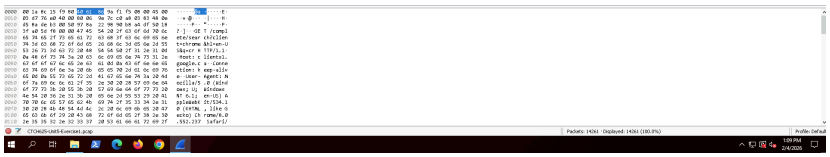


Figure 3: Sample PCAP file (2).pcap



5. CTCH 625 Unit 5 Exercise. PCAP Examination





Packet Number: 11

Time: 0.276759

Source IP Address: 192.168.3.131

Destination IP Address: 72.14.213.102

Protocol: TCP

Packet Length: 54

Source Port: 55950

Destination Port: 80

Sequence Number (Seq): 946

Acknowledgement Number (ACK): 488

Window Size (Win): 0

TCP Payload Length (Len): 0

Interpretation:

After reviewing the PCAP file in Wireshark, I examined packet number 11 in the capture list. The results show traffic from source IP address 192.168.3.131 to destination IP address 72.14.213.102 using the TCP protocol. The packet length is 54 bytes, and the communication is from source port 55950 to destination port 80, which is commonly associated with HTTP web traffic. The packet includes RST and ACK flags, which indicates the sender is resetting and closing the connection. The TCP session details show sequence number 946 and acknowledgment number 488. The window size is 0, indicating no more data is expected. Overall, packet 11 represents the termination of a TCP web connection between a client system and a web server, which is consistent with normal web browsing behavior.

6. Protocol Classification of Network Services and Applications

The PCAP file reviewed in Wireshark shows that packet 11 contains traffic using two main protocols, HTTP and TCP. These protocols support the network services and applications observed in the capture. A network protocol is a set of rules that defines how data is transmitted and received between devices on a network (GeeksforGeeks, 2026). These rules help ensure communication is consistent and reliable across different systems. In this capture, the protocol field helps identify what type of communication is taking place and which services and applications are involved in the network activity.

The Hypertext Transfer Protocol (HTTP) is associated with web services and is used for web page requests and responses between a client and a web server. In the packet capture, the HTTP protocol shows GET requests and server responses, which indicates normal web browsing activity and web content retrieval. HTTP is the foundation for data exchange on the web and operates using a client-server model (ByteByteGo, 2023). It allows browsers to request and load web pages while the server returns the requested resources. HTTP is not encrypted, so HTTPS is used when secure web communication is required.

The Transmission Control Protocol (TCP) is the transport layer protocol observed in the packet capture and is responsible for establishing and managing reliable connections between systems (Petrova, 2024). It is designed to send packets across the internet and ensure the successful delivery of data and messages (ByteByteGo, 2023). In the capture, the TCP packets show connection control behavior, and the web traffic sessions are indicated by acknowledgment and reset flags. TCP does not independently function as a user application but instead provides reliable data delivery for application protocols such as HTTP, FTP, and email services. This supports the web communication service observed in the packet capture.

Overall, the protocol types and port usage shown in packet 11 indicate that the traffic is mainly related to web communication. The capture shows HTTP packets with GET requests and server responses, followed by TCP packets using port 80 and connection control flags. This combination reflects normal web browsing network activity. TCP provides reliable transport for the connection, while HTTP handles the web content requests and responses. Based on the evidence, the main application involved is a web browser connecting to an external web server. This matches normal browser-based web access behavior.

7. Identify at least two network services and two applications

The Wireshark PCAP analysis shows that at least two network services and two applications can be identified from the packet 11 capture traffic. Network services are the core functions that enable communication and data exchange across systems (Brooks, 2025).

The first network service identified is web communication service, which is indicated by HTTP packets showing GET requests and server responses in the capture. HTTP operates using a client–server request and response model, where a client such as a web browser sends a request and the server returns the requested resource (Attari, 2024). GET requests are commonly used to retrieve web pages and related content, and these requests can be directly viewed in Wireshark

packet details because HTTP traffic is text-based (Mehta, 2023). The traffic is directed to destination port 80, which is used for standard web access. This is strong evidence that web content is being requested and delivered across the network.

The second network service identified in the packet capture is reliable transport service, provided by Transmission Control Protocol (TCP). TCP plays a critical role in network communication by establishing and controlling connections between systems and ensuring that transmitted data is received in the correct order and without loss (TechTarget, 2024). In the Wireshark capture, the TCP packets include acknowledgment and reset flags, which directly show that the protocol is managing the connection state and controlling when the session is closed. Packet 11 also displays sequence and acknowledgment numbers, which are used by TCP to track packet delivery and confirm receipt between the client and server. This packet-level evidence shows TCP actively managing reliability and session control while carrying the HTTP web traffic observed in the capture. TCP is not an end-user application itself, but a core network service that enables application protocols like HTTP to communicate reliably across the network.

After analyzing the two network services, application activity can also be identified from the packet capture. The first application identified is a web browser application. This is supported by the HTTP GET requests observed in the traffic. The capture shows the internal source IP address 192.168.3.131 sending requests to the external destination server 72.14.213.102. Web browsers generate GET requests when a user navigates to a website or requests online resources such as web pages, images, or scripts (GeeksforGeeks, 2025). In the Wireshark packet details, the readable HTTP request structure supports that a client-side browser is initiating the communication. Based on the timing and direction of the packets, the client system starts the session and waits for a server reply, which matches normal browser behavior. The use of destination port 80 further supports that this is standard web browsing activity. This request pattern and protocol behavior show that a browser-based client application is responsible for generating the observed packets.

The second application identified in the capture is a web server application. This is supported by the HTTP response packets that return from the external destination IP address 72.14.213.102 back to the client system after the GET request is sent. A web server application is responsible for receiving HTTP requests, processing them, and returning the requested resources to the client. In network traffic analysis, web servers send HTTP response messages that contain status codes and requested content in reply to client requests (Cisco, 2024). In the packet capture, the response behavior follows the client's GET request and shows the expected reply from the server. This confirms normal server-side handling of web requests. The request-and-response pattern provides clear evidence that a web server application is actively participating in the communication session observed in packet 11.

Overall, the packet 11 Wireshark PCAP analysis shows a complete and consistent web communication session supported by both network services and applications working together. The protocol evidence confirms that HTTP and TCP are operating together, with HTTP handling the web requests and responses, followed by TCP providing reliable transport, connection control, and ordered delivery. The presence of HTTP GET requests, server responses, port 80

traffic, and TCP acknowledgment and reset flags demonstrates a normal client–server web exchange. From this protocol behavior, two network services can be identified as web communication service and reliable transport service. The traffic pattern also identifies two applications, including a web browser acting as the client that sends the request, and a web server acting as the responder that returns the requested resources. The packet flow, protocol structure, and response sequence align with standard browser-to-server web activity. When the packet evidence is combined, it shows that the traffic represents normal and properly functioning web communication across the network.

Exercise 2: Categorize application protocols using common network models.

The client–server model is a common network model used by application protocols. It is designed to support secure and efficient communication between systems. Each of these protocols demonstrates this structure through HTTP, HTTPS, WebSocket, and SMTP, while following a defined communication pattern. Each plays a specific role in maintaining reliable and secure network operations.

The Hypertext Transfer Protocol (HTTP) operates using a client–server network model where a client, typically a web browser, sends requests and a web server returns responses. It is the foundation of web communication and is used to retrieve web pages and related resources. Each HTTP request is transmitted over the internet as structured and encoded data containing different types of information. The request usually contains an HTTP version, a URL, an HTTP method, HTTP request headers, and an optional HTTP body (Cloudflare, n.d.). HTTP supports efficient communication by using a standardized request and response format, which allows different systems and platforms to exchange data consistently. However, HTTP itself is not encrypted, which means data can be exposed if additional security controls are not used (ByteByteGo, 2023). When used independently, it lacks built-in encryption and creates security risk.

Hypertext Transfer Protocol Secure (HTTPS) uses the same client–server model as HTTP but adds Transport Layer Security (TLS) encryption to protect communications. HTTPS is important for secure network communication. By encrypting data in transit, it protects confidentiality and helps verify server identity through digital certificates. This reduces the risk of interception and tampering during transmission. By protecting data, it allows encrypted communications between internal and external services and secure transmission channels to prevent interception, safeguard traffic through robust website authentication mechanisms, and prevent advanced phishing tactics (SecurityScorecard, 2025). HTTPS maintains efficiency while adding security controls, which is why it is now the standard protocol for secure web traffic (ByteByteGo, 2023). As an essential security protocol for FIC Bank, it helps protect login credentials, financial data, and personal information transmitted between users and web servers.

WebSocket is another protocol that uses a client–server model but is designed for persistent, full-duplex communication over a single TCP connection. It keeps the connection open, allowing for real-time, two-way communication, which makes it well suited for live chats and online games where consistent updates are needed (GeeksforGeeks, 2025). Unlike traditional HTTP request-response behavior, WebSocket allows servers to push updates to clients in real time without

repeated polling. This makes it efficient for organizations such as FIC Bank that use applications requiring financial dashboards. By maintaining an open connection, it improves performance and reduces overhead, but it must be secured through encryption and proper authentication controls to prevent misuse (ByteByteGo, 2023).

Simple Mail Transfer Protocol (SMTP) also follows a client–server model and is used to send email between mail servers and clients. Properly configured SMTP services help maintain reliability and message integrity in organizational communication systems (ByteByteGo, 2023). It is important for efficient message transfer because it provides a standardized method for how email is routed and delivered across networks. From a security perspective, SMTP can be combined with encryption extensions and authentication controls to reduce spoofing and interception risks.

Overall, these protocols demonstrate how application protocols are categorized within client–server network models and how each supports secure and efficient communication in different ways. HTTP and HTTPS support web content delivery, WebSocket supports real-time bidirectional communication, and SMTP supports structured email transfer. Their defined communication models, combined with encryption and control mechanisms, help maintain both performance and security across FIC Bank networks.

Exercise 3: Interpret network traffic and monitor data.

Part 1: Capturing YouTube Video Streaming Traffic

Figure 1: Wireshark filtered TLS/ 10.11.122.212 (My System)

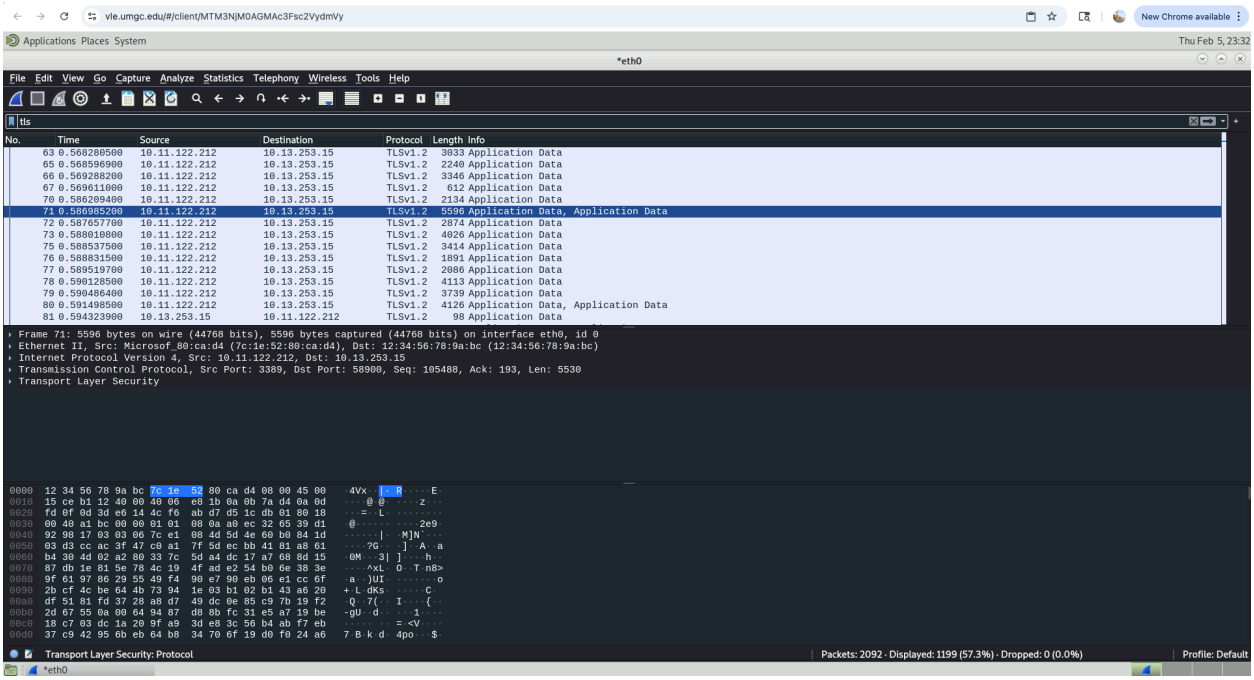
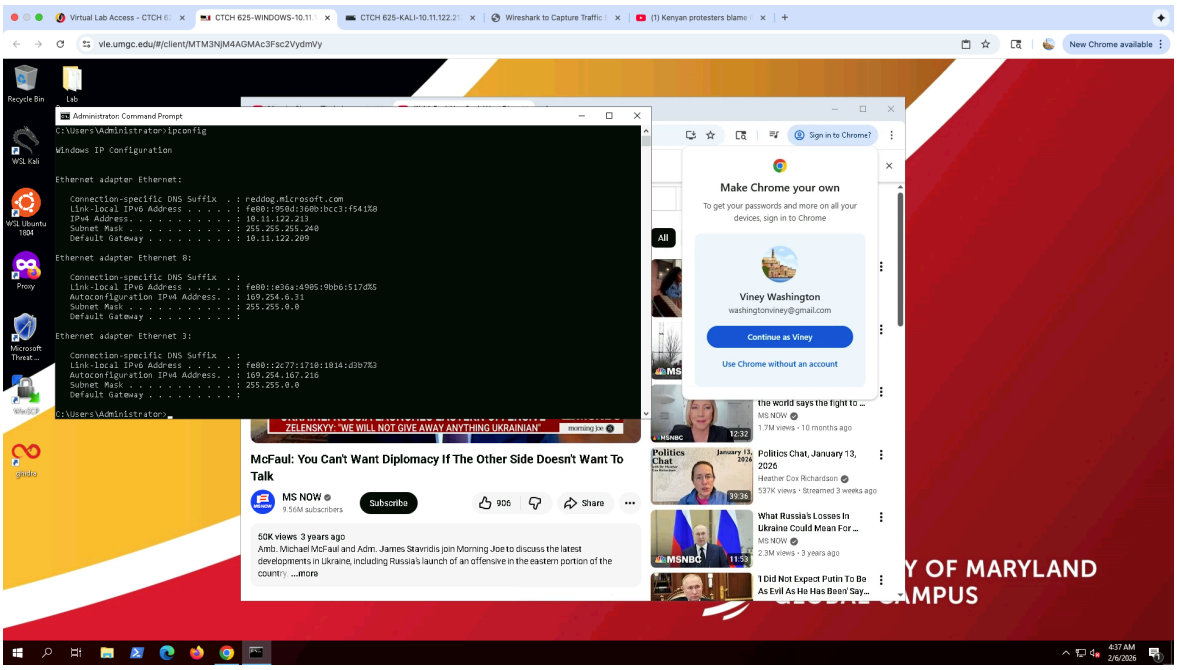


Figure 2: 10.13.253.15 (YouTube Server)



Part 2: Capturing an Incoming Connection

Figure 1: Windows IP Address 10.11.122.213

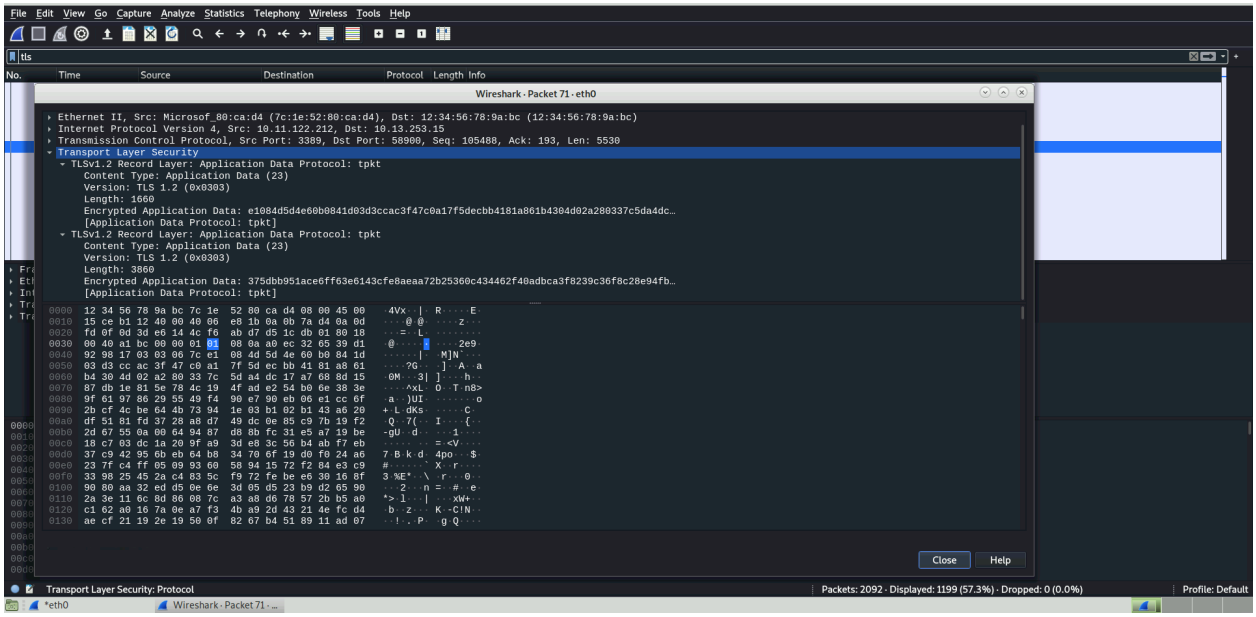


Figure 2: Kali Terminal Ping (10.11.122.213)

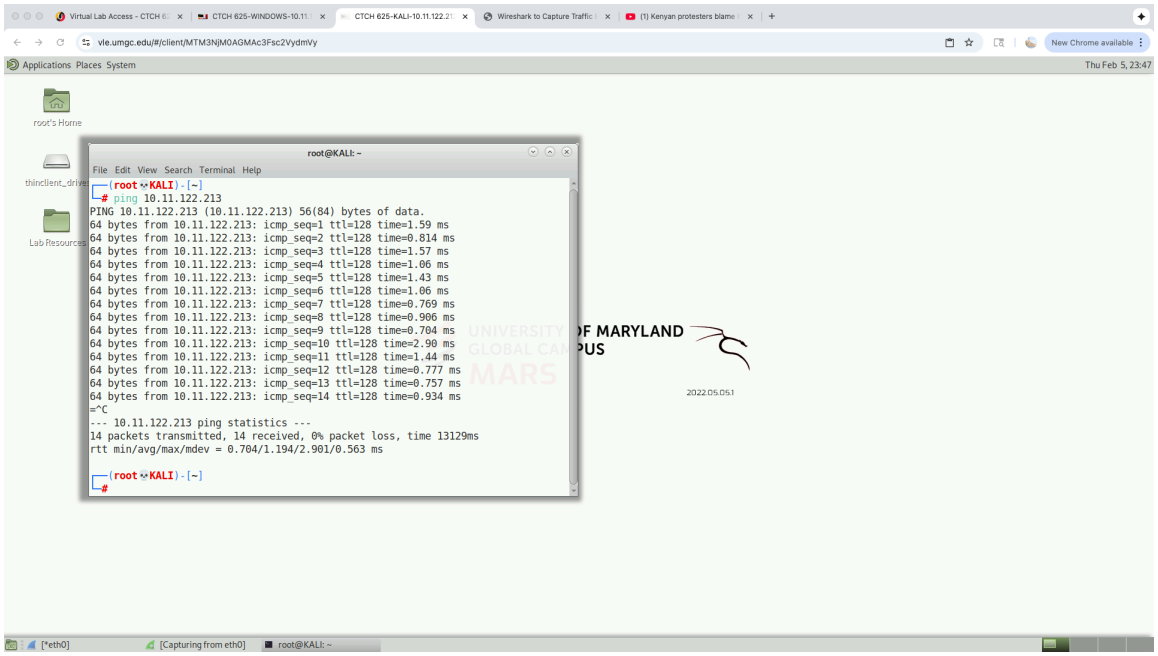


Figure 3: Wireshark -Filter ICMP (10.11.122.213)

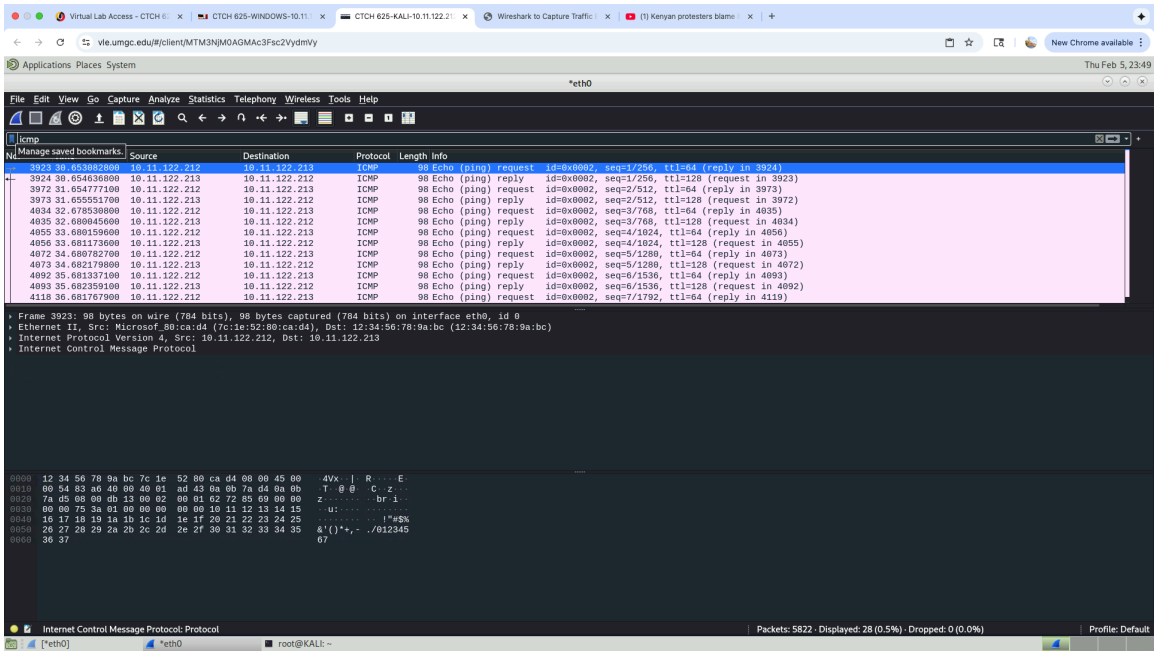


Figure 3:Internet Protocol Version 4

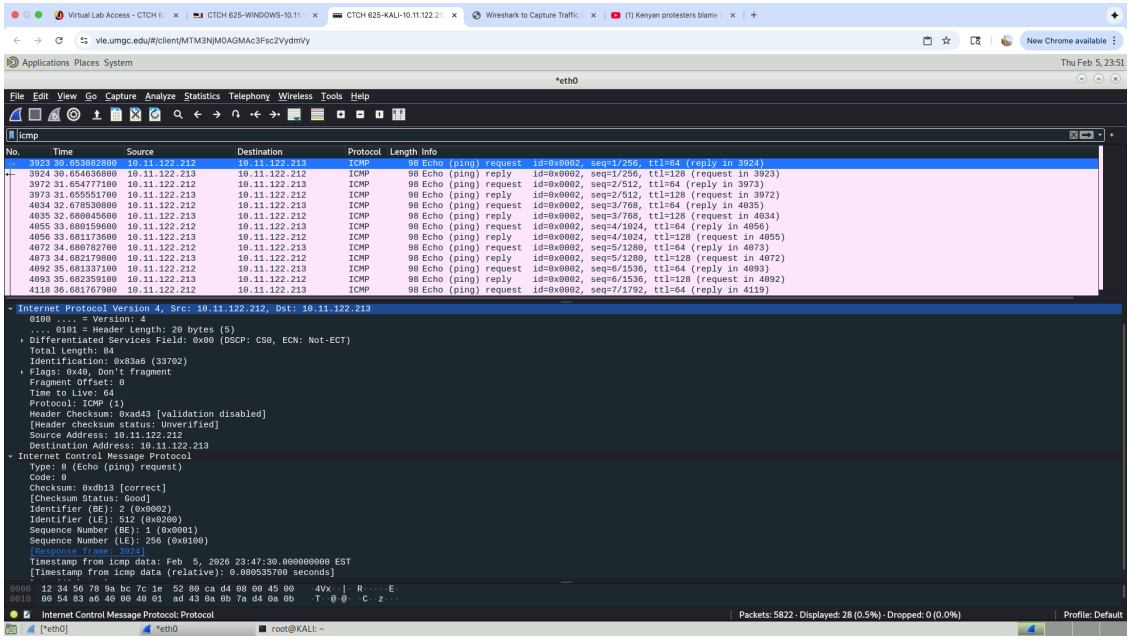


Figure 4:Interne Control Messaget Protocol

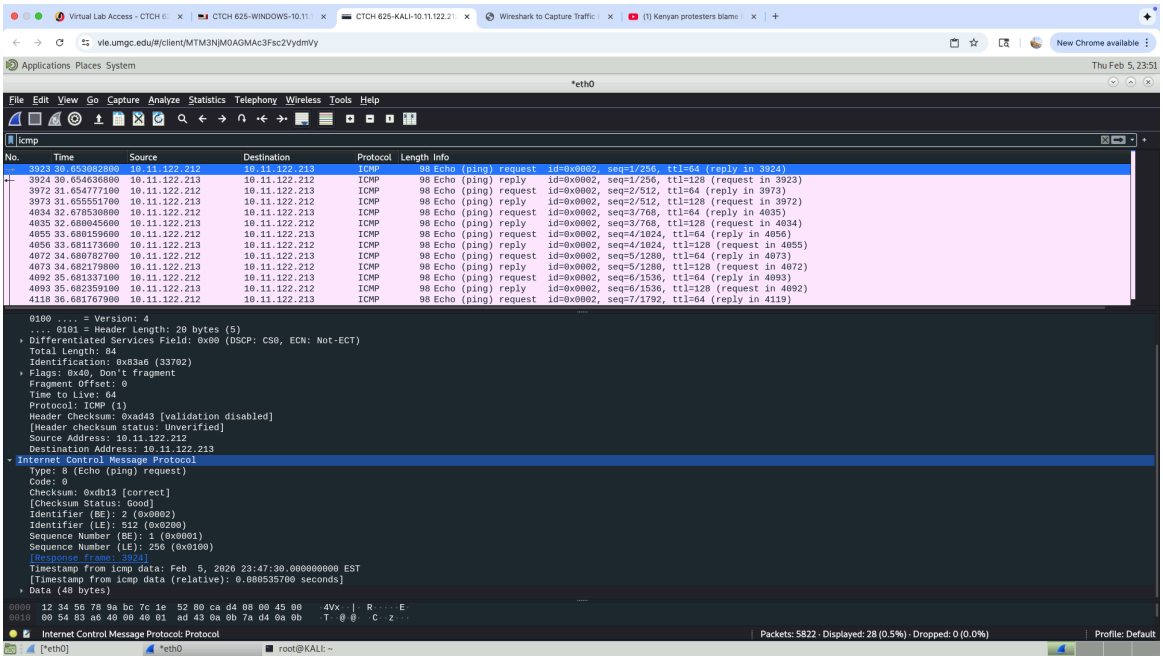


Figure 5: Packet list with ICMP rows Source & Destination IP columns ICMP request/reply lines visible Lower detail pane open

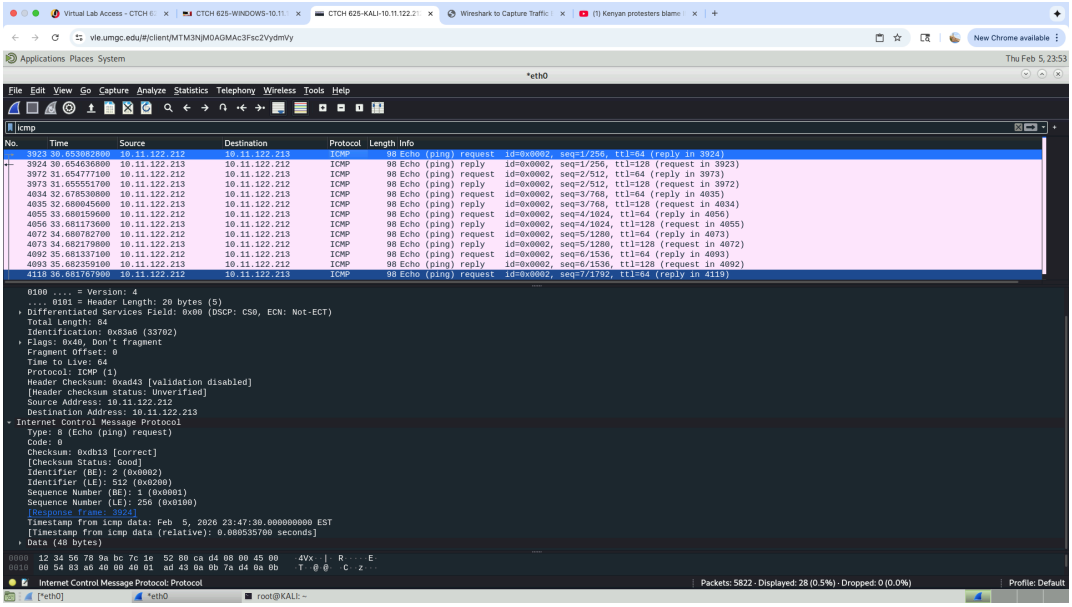
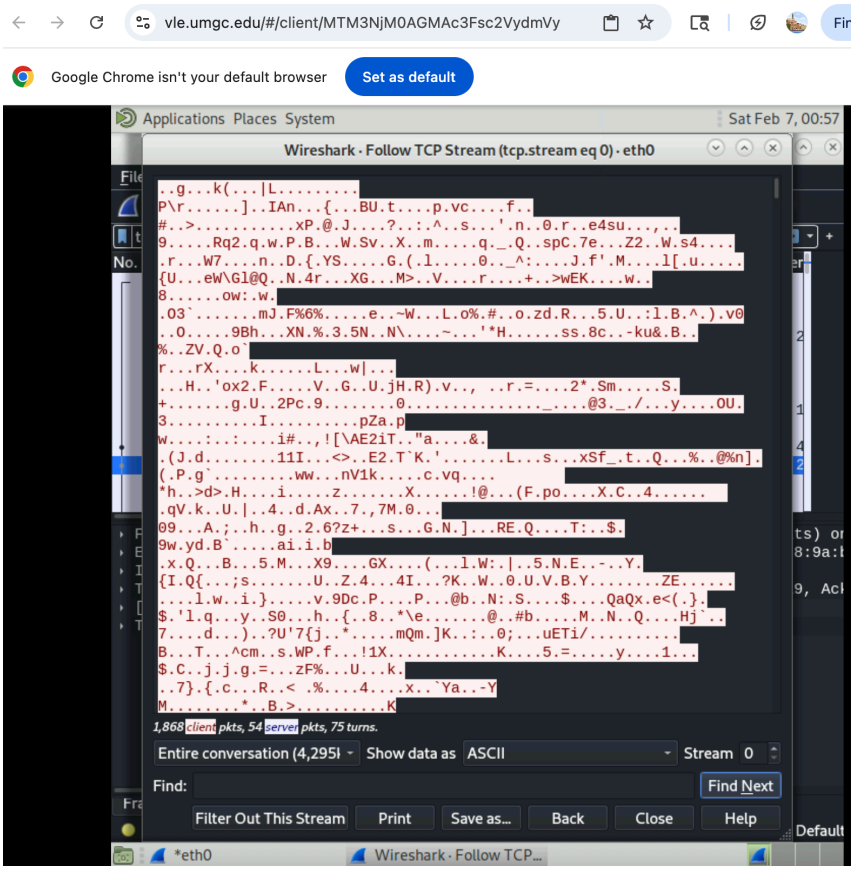


Figure 6: TCP Stream Results



Analyze Captured Packets for Unusual Traffic Violating FIC Bank Policy

The Wireshark packet captures from Part 1 and Part 2 show mostly normal client server communication patterns. Depending on FIC Bank's firewall security policies, they demonstrate traffic types that could be considered unusual or policy violations. The analysis focuses on TLS encrypted web traffic and ICMP diagnostic traffic using protocol filters and packet inspection.

In Part 1, the YouTube streaming test was captured using the TLS filter. The packet list shows repeated TLSv1.2 Application Data packets between the internal host 10.11.122.212 and the external server 10.13.253.15. Multiple frames show large packet sizes in the several thousand byte range, which is consistent with encrypted video streaming behavior rather than short web page requests. TLS is used to secure HTTPS communications by protecting confidentiality and integrity of transmitted data. The TLS record protocol identifies message types such as handshake and application data and verifies message integrity during encrypted transmission (Heetmehta, 2025). The traffic pattern shows evidence of normal encrypted HTTPS client server communication, but under FIC Bank policy, continuous encrypted streaming to non business services such as video platforms could be classified as unusual or non compliant usage. Financial organizations commonly restrict streaming services because they create long lived outbound sessions and high bandwidth consumption. While TLS encryption protects data in transit, it also reduces packet content visibility unless inspection controls are deployed (Cisco, 2024).

In this capture specifically, the TLS stream identified repeated application data packets between the same internal and external IP address pair across a sustained time period instead of short request response exchanges. That sustained encrypted session pattern is a key indicator that the traffic represents media streaming instead of standard transactional web browsing. From a FIC Bank policy perspective, this could be flagged as inappropriate network use inside a banking environment even though the protocol itself is secure.

In Part 2, the capture filtered with ICMP shows diagnostic ping traffic between internal systems. The ICMP packet list displays repeated echo request and echo reply packets between 10.11.122.212 and 10.11.122.213, with sequential identifiers and matching reply frames. When pinging a host, systems may return echo replies or ICMP error messages such as destination unreachable or time exceeded depending on filtering behavior (Gowthamra, 2024). The Kali terminal output confirms successful ping responses with zero packet loss, which matches the ICMP reply packets observed in Wireshark. ICMP is commonly used for troubleshooting and reachability testing, but security guidance explains that ICMP can also be abused for reconnaissance and host discovery by attackers mapping network systems.

In this specific capture, the pattern shows multiple ICMP echo requests, matching echo replies, the same source and destination IP pair repeating, visible sequence numbers, and successful round trip behavior. This indicates authorized diagnostic testing activity. Under FIC Bank security policy, repeated ICMP sweeps across many hosts instead of a single approved test target would be considered unusual traffic and a likely policy violation. Firewall policies should only permit appropriate source and destination IP addresses to be used (Scarfone and Hoffman, 2009).

The TCP stream view from the capture also confirms active session behavior tied to encrypted web traffic. The Follow TCP Stream window shows a long client and server conversation with many packets and turns, confirming a sustained session instead of a failed or blocked connection. The stream contains continuous bidirectional packet flow between 10.11.122.212 and 10.13.253.15, which indicates a successfully established and maintained TCP session rather than a denied or filtered connection. In firewall analysis, unusual TCP behavior such as repeated failed handshakes, unexpected resets, or abnormal port access can indicate blocked or malicious activity. When firewall controls are working correctly, blocked sessions typically appear as incomplete handshakes, missing server replies, or immediate reset responses instead of sustained packet exchanges. TCP is designed to establish reliable sessions through handshake and acknowledgment sequences, so abnormal patterns can signal filtering or misuse (ElastiFlow, n.d)

Overall, the captured packets show TLS encrypted streaming traffic and ICMP diagnostic traffic, but when evaluated against FIC Bank security policy, continuous video streaming sessions and unrestricted ICMP testing could be classified as unusual or restricted activity. The specific packet evidence includes repeated TLSv1.2 application data frames between 10.11.122.212 and 10.13.253.15 with large byte sizes and sustained TCP stream data, along with repeated ICMP echo request and reply pairs between 10.11.122.212 and 10.11.122.213 with matching identifiers, sequence numbers, and zero packet loss. These packet level indicators demonstrate that access controls allowed the tested traffic types, which is appropriate for lab diagnostics but would require tighter restriction in a production banking environment. Blocking or limiting non business streaming domains, restricting ICMP to approved administrative hosts, and monitoring long duration encrypted sessions for policy compliance are measurable firewall control actions that can be validated through follow up packet capture testing, which aligns with firewall best practice guidance for protected financial networks.

3. Examples of Unusual traffic, and Potential Remedies.

FIC Bank considers unusual network traffic to be any packet behavior or communication pattern that does not align with approved business or operational network use within the organization. In a secure financial environment, traffic is expected to follow approved protocols, approved destinations, and defined usage patterns. When those patterns are violated, packet captures often show repeated, excessive, or unauthorized communication behavior that can indicate policy violations or elevated security risk.

One example of unusual traffic is a high volume of encrypted requests coming from a single IP address or a small group of IP addresses (Nuutan, n.d.). A sudden spike in encrypted traffic volume may indicate a Distributed Denial of Service event or abnormal internal system behavior. In the packet capture, repeated TLS application data packets and long duration encrypted sessions to an external server indicate sustained media streaming behavior. While TLS encrypted HTTPS traffic is normally expected, long running outbound encrypted sessions to non-business platforms such as YouTube would be considered non-compliant traffic in a banking network. Financial organizations commonly restrict streaming services because they create long lived outbound sessions and high bandwidth consumption. Abnormal outbound traffic volume and unexpected external communications are recognized indicators of policy misuse or elevated

threat activity (SentinelOne, 2025). A practical remedy is to apply firewall and web filtering controls to block or limit non-essential streaming domains and enforce acceptable use through outbound traffic rules. Firewalls monitor incoming and outgoing traffic for abnormalities and function as a first line of defense. Additional controls may include IP reputation blocking, domain filtering, and traffic rate thresholds to automatically restrict excessive encrypted session behavior (Nuutan, n.d.).

A second example of unusual traffic is repeated ICMP scanning or ping sweep behavior across multiple internal hosts. Security monitoring systems may trigger alerts when packet sizes or ICMP frequency levels appear abnormal for standard Windows or Linux platform behavior (Plixer, n.d.). In the capture, ICMP echo requests and replies were observed between two approved lab systems, which is acceptable for diagnostics. However, if the same ICMP request pattern appeared across many destination hosts instead of a single authorized target, it would indicate reconnaissance or host discovery behavior. ICMP misuse can support mapping attacks and may be used as a smokescreen for other malicious actions such as unauthorized access attempts (Blue Goat Cyber, n.d.).A recommended remedy is to configure firewall rules to allow ICMP only from approved management systems and to rate limit ICMP traffic volume. Setting packet rate thresholds and restricting ICMP response scope reduces reconnaissance exposure while preserving diagnostic capability (Blue Goat Cyber, n.d.).

Another example of unusual traffic is abnormal TCP session behavior, such as repeated failed connection attempts, incomplete handshakes, or unexpected reset responses across sensitive ports. TCP is designed to complete structured handshake and acknowledgment sequences for valid sessions. When packet captures show repeated SYN attempts without successful completion or frequent reset responses, this can indicate blocked intrusion attempts or unauthorized port probing (Yasar, 2024). Remedies include tightening inbound firewall rules, restricting exposed ports, and enabling alerts for repeated failed connection attempts. TCP session validation controls such as SYN checks and sequence verification can also be enabled at the firewall policy level to ensure sessions follow expected behavior patterns (Juniper, 2023).

Overall, unusual traffic is identified by patterns that differ from expected business communication, including sustained non business encrypted sessions, scanning behavior, and abnormal session control patterns. Effective remedies include firewall rule enforcement, outbound filtering, protocol restrictions, rate limiting, and continuous packet monitoring. These remedies are directly supported by packet capture evidence and align with financial sector firewall and network security best practices.

Exercise 4: Estimate risk associated with web application protocols.

There is a severity of risk for web applications when using the OWASP Risk Rating methodology. The severity of risk for web applications is estimated by evaluating both likelihood and impact. The OWASP explains the core model of risk equals likelihood and multiple by impact for application security. Likelihood measures the chances that a vulnerability can be discovered and exploited, while impact measures the level of technical and damage if exploitation occurs. The purpose of this model is to “estimate the severity of all of these risks to the business and make an informed decision about what to do about those risks” (Williams, n.d.).

Applying this structured method to different web applications used in FIC Bank will identify how risk severity changes based on exposure, attacker opportunity, and business consequences.

An online banking login application would be rated as critical risk. The authentication portals are publicly reachable and constantly targeted rating the likelihood as high. The likelihood is “how likely this particular vulnerability is to be uncovered and exploited by an attacker” (OWASP, n.d.). Modern web threat guidance confirms that access control failures are among the most severe risks and states. Broken access control occurs when an application fails to properly restrict what authenticated users are allowed to do, allowing attackers to access unauthorized functionality and data” (StackHawk, 2025.). The impact is high because it expose financial accounts and personal identity date, if exploits is successful. OWASP severity mapping places high likelihood and high impact in the critical category. This supports controls such as multi factor authentication, strict access control enforcement, and least privilege permissions.

An internal employee dashboard would be rated as high risk. The likelihood is lower than a public site, but internal and authenticates users are valid threat agents. OWASP specifically instructs testers to “use the worst-case threat agent” when estimating likelihood (OWASP, n.d.). The impact is high due to the employee dashboards that contain operational records and customer information. Access control failures are especially dangerous in these systems because attackers can move from normal access into unauthorized data areas. In modern web security guidance, “An attacker could potentially access other users’ account information by changing the account number in the URL” (StackHawk, 2025.). Business impact is high, OWASP scoring results in high severity even when likelihood is only medium. This justifies role based access controls, audit logging, and session monitoring.

A customer support chat or service portal would be rated as medium risk. The likelihood is medium to high since the application is public facing and accepts user input, which increases exposure to misuse and automated attacks. OWASP defines likelihood as “how likely this particular vulnerability is to be uncovered and exploited by an attacker” (OWASP, n.d.). The impact is typically medium, since chat and support portals do not usually provide direct access to core banking functions, but they can still expose customer information or be abused as an entry point for additional attacks. When monitoring controls are weak, risk increases and attacker activity may go undetected. As security guidance explains, “Insufficient logging and monitoring refer to a lack of proper logging and monitoring processes to detect and respond to security threats” (StackHawk, 2025). When medium likelihood and medium impact are combined, the OWASP severity scale identifies this type of application as medium risk. This supports protections such as input validation, strong session controls, comprehensive logging, and real time alerting.

A public marketing website would be rated as medium risk under the OWASP Risk Rating Methodology. The likelihood is high since the site is fully public and easily reachable by anonymous internet users. when estimating attacker size and access level, the likelihood measures “how likely this particular vulnerability is to be uncovered and exploited by an attacker” (OWASP, n.d.). The impact is typically lower than internal or banking transaction systems since marketing sites usually do not store sensitive financial records or provide account

access. The most likely damage would be website defacement, misinformation, or reputation harm instead of direct account compromise. By monitoring gaps increases exposure time.

Insufficient logging and monitoring refer to a lack of proper logging and monitoring processes to detect and respond to security threats" (StackHawk, 2025). When high likelihood is combined with lower business impact, the OWASP severity scale places this type of application in the medium risk category. This supports applying practical protections such as regular patching, secure configuration, and continuous logging and monitoring controls.

Overall, applying the OWASP Risk Rating Methodology shows that web application risk is not judged by exposure alone, but by the combination of likelihood and business impact. Public facing authentication systems such as online banking login portals fall into the critical category since they are heavily targeted and a successful exploit would cause severe financial and privacy damage. Internal employee dashboards rank as high risk since the likelihood is lower but the business and data impact remains high if access controls fail. Customer support portals and public marketing websites fall into the medium risk range since they are easier to reach and test, but usually have lower direct financial impact, although they still create customer data and reputation risk if abused. The OWASP model supports consistent decision making by tying severity to measurable likelihood and impact factors, which helps organizations like FIC Bank prioritize stronger controls, monitoring, and protections on the applications where compromise would cause the greatest harm (OWASP, n.d.).

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